Wikispeedia Network Analysis

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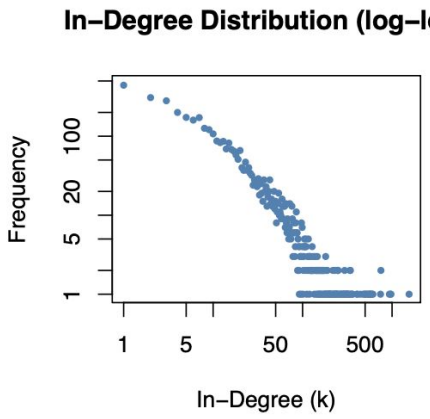
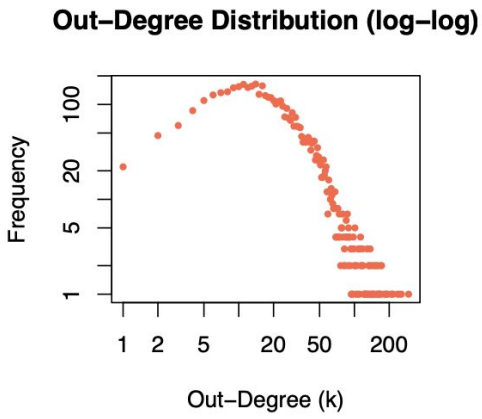
What is Wikispeedia?

- Wikispeedia is an online game where users attempt to navigate from one Wikipedia article to another using only hyperlinks embedded in the articles.
- The goal is to start from Article A and finish at Article B in as fewest clicks as possible.
- We model it as a directed network articles are nodes and hyperlinks are edges
- This particular dataset uses a smaller version of Wikipedia with only 4,592 nodes (articles) and 119,000+ edges (hyperlinks)

Network Structure

- Diameter = 9, average path length = 3.18 → small-world network
- Reciprocity = 0.221 → 22% of links are mutual
- Clustering coefficient = 0.103
- In-degree is heavy-tailed: a few mega-hubs, most articles barely linked
- The out-degree distribution is somewhat less skewed, since the number of outgoing links depends more on how detailed an article is.





Network Visualizations

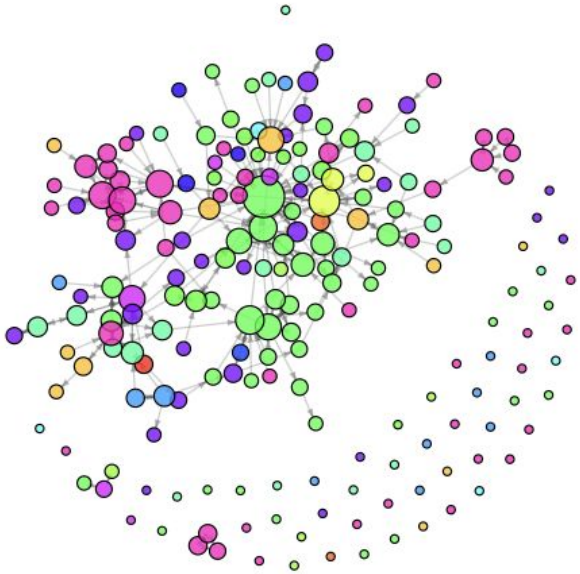
- We sampled 4 random subgraphs of 200 nodes each from the full network
- Nodes are colored by Wikipedia category and sized proportionally to their degree
- Three layout algorithms used for each subgraph:
  - **Fruchterman-Reingold**: force-directed, pulls high-degree hubs to the center
  - **Kamada-Kawai**: positions nodes so graph distances match geometric distances (# of clicks)
  - **Circular**: arranges all nodes in a ring, making edge density easy to see
- Across all layouts: well-connected hub nodes cluster at the center, low-degree nodes pushed to the periphery
- Circular layout makes it clear edges are not evenly distributed, some regions are far more densely connected

Network Visualizations

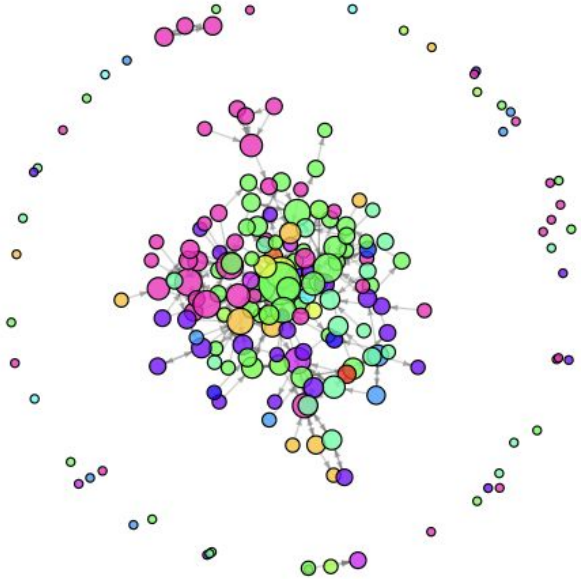
Node Category

- Art
- Business_Studies
- Citizenship
- Countries
- Design_and_Technology
- Everyday_life
- Geography
- History
- IT
- Language_and_literature
- Mathematics
- Music
- People
- Religion
- Science
- Uncategorized

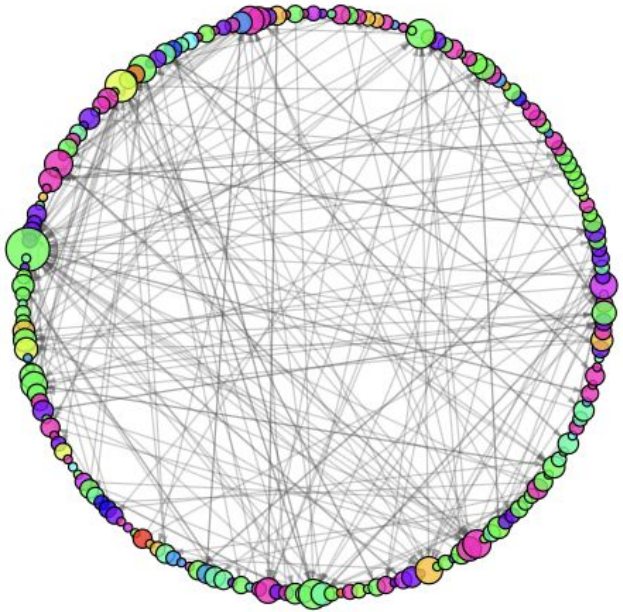
Subgraph 1 – Fruchterman–Reingold



Subgraph 1 – Kamada–Kawai



Subgraph 1 – Circular



Most Influential article:

The “United_States” article is the most central article in the Network

In Degree (popularity):

- 1,551 incoming links
- Most linked-to article

Betweenness (Bridge role):

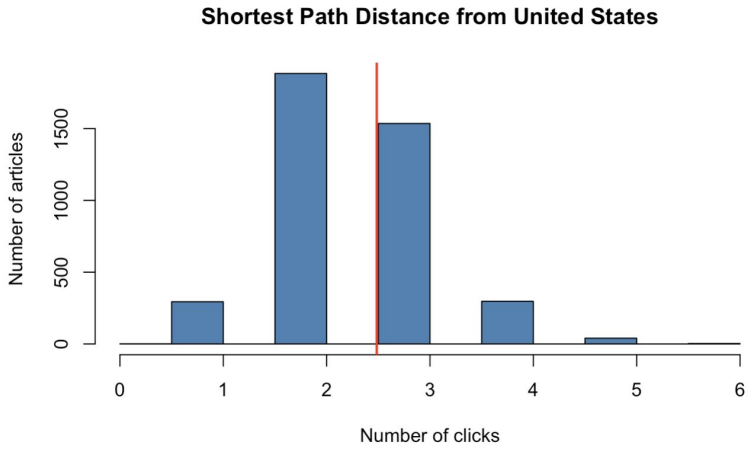
- Appears on the largest number of shortest paths
- Connects many different topics

Eigenvector Centrality (influence)

- Score = 1.000 (highest possible)
- Linked by other important articles

Most articles are only 2-3 clicks away from the United States

- Average distance is 2.48



Community Detection

- Used three different community detection algorithms
- Louvain has the highest modularity, making it our best algorithm
- Strong evidence because algorithms have similar communities

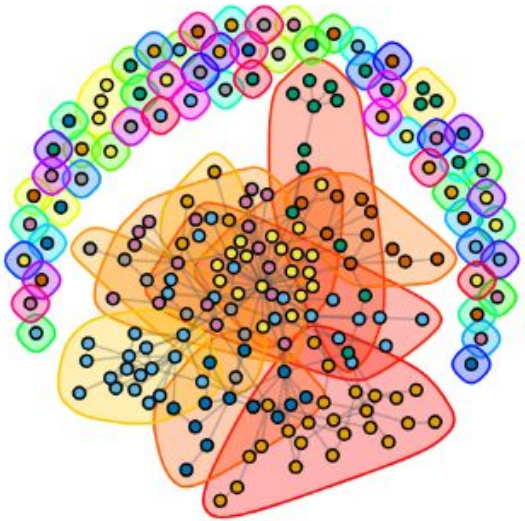
Algorithm	Communities	Modularity
Fast Greedy	68	0.6611
Louvain	68	0.6634
Walktrap	77	0.6147

- Pre-made Categories tend to be in the same Communities.

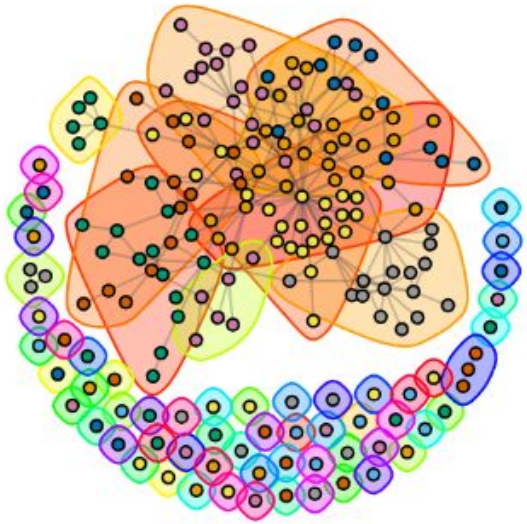
Community	Size	Top Categories (count)
Community 4	23	Everyday_life (6), Science (5), Geography (4)
Community 7	21	Geography (10), People (4), Citizenship (2)
Community 8	17	Science (13), People (3), Citizenship (1)
Community 3	14	People (5), History (3), Geography (2)
Community 6	14	People (4), Citizenship (3), Everyday_life (3)

Community Detection Visualization

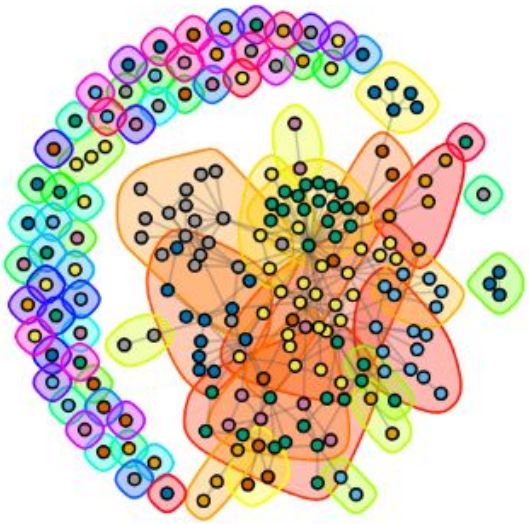
Fast Greedy (Q=0.661)



Louvain (Q=0.663)

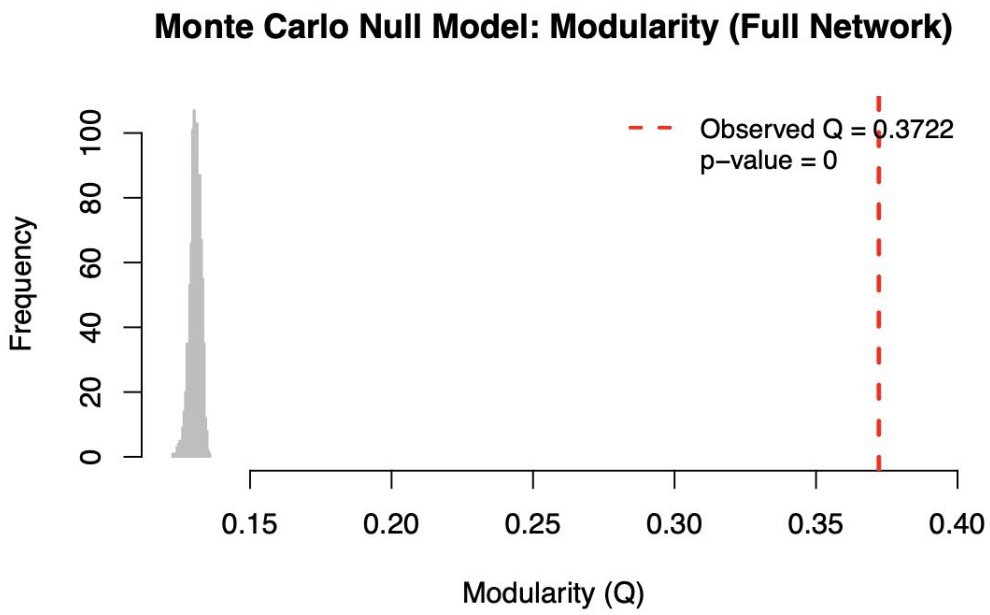


Walktrap (Q=0.615)



Statistical Validation: Is the community structure real or just random wiring?

- Methodology: Compared our Louvain communities against 1,000 randomly wired networks.
- Results: Our network's modularity (Q=0.3722) vastly outperformed the random distribution (p≈0).
- Conclusion: The network's community structure have real underlying organization with Wikipedia's categories and are not just random chance.



Results & Conclusion

- Key Findings: Network Structure and Centrality
  - The network has a short diameter of 9 and an average path length of 3.18 which explains why Wikispeedia is easily played.
  - About 22.1% of hyperlinks are mutual which shows a lot of cross-referencing between related articles.
  - Degree and centrality metrics show that a small core of broad topics act as major hubs while a few high betweenness edges act as bridges between different categories.
- Community Detection & Significance
  - Our three algorithms consistently found communities that aligned with the actual Wikipedia subject categories.
  - Our Monte Carlo null model test of 1,000 random graphs confirmed that the observed community structure is highly significant and not just by chance ($Q = 0.3722$, $p = 0$).
- Future Direction
  - Next step would be to incorporate actual player data to build a weighted network and shift the focus from static hyperlinks to actual human navigation behavior.

wikispeedia
game link